

LOGISTIC REGRESSION

Module 3: Predicting the Future

1. THE CLASSIFICATION PROBLEM

Linear regression is designed to predict values that can range from $-\infty$ to $+\infty$. But what if the answer is just binary?

- *Example:* Will a customer click this button? (1 = Yes, 0 = No)
- *Example:* Is this image a cat or a dog? (1 = Cat, 0 = Dog)

If you use a straight line here, you get nonsense probabilities (like -150% or 400%). We need a way to squash the output to stay strictly between **0 and 1**.

2. THE SIGMOID FUNCTION

To solve this, we pass the output of our linear equation through the **Sigmoid Function** (also known as the Logistic Function). It takes any input and “squashes” it into a smooth, S-shaped curve.

The Sigmoid Formula:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- As $z \rightarrow \infty$, the output $\sigma(z) \rightarrow 1$.
- As $z \rightarrow -\infty$, the output $\sigma(z) \rightarrow 0$.
- At $z = 0$, the output is exactly **0.5**.

3. THE LOG-ODDS RATIO

Logistic regression doesn't predict the probability directly; it predicts the **Log-Odds** (the logit).

- **Odds:** The ratio of success to failure: $\frac{p}{1-p}$.
- **Logit:** We take the natural log of the odds to turn the S-curve back into a linear relationship, allowing our computer to solve for the best fit.

$$\ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X$$

This allows the algorithm to learn the optimal parameters (β) that move the S-curve left, right, or steeper, until the 0.5 probability threshold perfectly separates the two categories.

4. THE DECISION BOUNDARY

Because our Sigmoid output is always a probability (between 0 and 1), we need a **Decision Boundary** to make a final call.

- Standard practice is the **0.5 threshold**.
- If $\sigma(z) \geq 0.5$, classify as **1** (Yes).
- If $\sigma(z) < 0.5$, classify as **0** (No).

You can shift this threshold depending on your priorities. For example, if you are screening for a deadly disease, you might set the threshold at **0.2**, choosing to classify more people as "likely sick" to ensure you don't miss any true cases (prioritizing Recall over Precision).

5. BEYOND BINARY

The logic of Logistic Regression is the foundation of **Neural Networks**. A single neuron is basically just a Logistic Regression unit. By stacking these units together, we can classify the entire universe.